Mental Health Crisis Early Warning System for Campuses

SYNOPSIS OF THE PROJECT

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1. ABSTRACT

University students often experience substantial changes in sleep, physical activity, social engagement, daily routine, and technology use during periods of academic pressure, isolation, transition, or emotional difficulty. Traditional campus mental-health services commonly depend on students recognizing a problem and voluntarily contacting a counsellor, which may delay support. This project proposes an ethical, opt-in Mental Health Crisis Early Warning System for university campuses that uses privacy-conscious passive sensing and machine learning to identify changes in wellbeing-related behaviour and encourage timely, human-centred support. The proposed platform is not intended to diagnose depression, anxiety, or any other psychiatric disorder. Instead, it detects patterns that may indicate a deterioration from a student's own recent baseline and converts those patterns into a wellbeing-risk indicator. The proposed system uses a Python-based technology stack. A student-facing application or web/mobile interface collects consent and, where technically available and permitted, receives limited behavioural signals such as sleep duration or regularity, activity level, device-use patterns, and coarse measures of social interaction frequency. For a college-level prototype, real device data can be replaced by simulated or voluntarily entered data. The backend is implemented using FastAPI, with SQLite for the prototype and PostgreSQL as a possible production database. Pandas and NumPy support preprocessing and feature engineering, while Scikit-learn can be used to train and evaluate machine-learning models such as Logistic Regression and Random Forest. A longitudinal baseline is maintained for each consenting participant so that the system focuses on meaningful changes rather than comparing every student with a single population norm. The system produces a risk score and risk band such as low, moderate, or high wellbeing concern. When a configurable threshold is reached and sufficient data quality is available, the student receives a confidential, non-diagnostic wellbeing check-in and links to appropriate campus resources. Counsellors receive a prioritised list based on risk and persistence, subject to institutional policy and the student's consent. Administrators receive only privacy-preserving aggregate analytics, such as campus-level trends and percentages, rather than unnecessary individual information. Security measures include role-based access, encryption in transit, pseudonymous identifiers, data minimisation, retention limits, audit logging, and the ability to withdraw consent. The project follows an iterative software-development and machine-learning methodology: requirements and ethics analysis, consent-aware data collection, preprocessing, feature engineering, model development, validation, system integration, usability testing, and evaluation. The expected outcome is a working prototype demonstrating how passive sensing, machine learning, and privacy-by-design can support earlier and more personalised wellbeing outreach while keeping human counsellors responsible for interpretation and intervention.

Keywords: mental health, digital phenotyping, passive sensing, machine learning, student wellbeing, early warning, privacy, consent, FastAPI, Python.

2. INTRODUCTION

Mental health and wellbeing have become important concerns within university environments because students move through a period of rapid academic, social, financial, and personal change. University life can involve new living arrangements, separation from family, demanding coursework, examinations, employment, relationship changes, social adjustment, uncertainty about careers, and pressure to perform. These experiences do not automatically constitute mental illness, but they can be associated with changes in sleep, physical activity, social connection, concentration, daily routine, and technology use. A technology system that can identify meaningful changes in these everyday patterns may therefore help universities offer support before a student reaches a point at which a crisis becomes severe.

Most conventional campus mental-health workflows are reactive. A student may first notice that something is wrong, decide that professional help is appropriate, find the counselling service, make an appointment, and disclose enough information for a counsellor to understand the situation. Each step requires motivation, time, awareness, and trust. Students who are struggling may not have the energy or confidence to initiate contact. Universities also face practical limitations: counselling services have finite staff capacity, and counsellors cannot continuously observe the wellbeing of every student. A complementary early-warning platform could help staff focus attention on students who have opted into proactive support while preserving the central role of human professionals.

Recent research in digital phenotyping provides a technical basis for this idea. Digital phenotyping generally refers to the use of data generated by smartphones, wearables, and other digital devices to quantify patterns of behaviour and context. Smartphone sensors can provide information related to movement, mobility, device interaction, and communication patterns, while wearables can provide information related to activity and sleep. The StudentLife study demonstrated that smartphone sensing could capture associations among sleep, activity, sociability, stress, mood, and academic outcomes in college students. Later research has continued to explore how passive sensing can be related to depression, anxiety, stress, and daily mood.

The value of passive sensing is that it can reduce the burden of repeatedly asking a student to complete long questionnaires. Instead of relying only on retrospective self-report, a system can observe changes in behaviour with the user's permission. However, passive data should never be treated as a direct measurement of a mental disorder. A student may sleep less because of examinations, travel, employment, cultural events, illness, or personal preference. Reduced social interaction may reflect studying, disability, introversion, or temporary circumstances. Increased phone use may be associated with many benign activities. Consequently, the proposed project treats passive signals as indicators of possible wellbeing change rather than clinical diagnoses.

A central design principle of the proposed project is therefore the use of a personal baseline. Instead of deciding that five hours of sleep is automatically high risk for every student, the system can compare a student's recent pattern with their own historical pattern. For example, if a consenting student normally sleeps approximately seven and a half hours but changes to four to five hours for several consecutive days while also showing reduced activity and reduced social interaction, the combined deviation may be more informative than any single signal. This approach also makes the system more personalised and reduces dependence on a universal definition of normal behaviour.

Machine learning can support the transformation of multiple behavioural features into a wellbeing-risk indicator. The project can begin with interpretable models such as Logistic Regression and Decision Trees and then compare them with ensemble methods such as Random Forest or Gradient Boosting. Model selection should be based not only on predictive performance but also on interpretability, calibration, robustness to missing data, and fairness. In a mental-health context, a model that is slightly less accurate

but easier to understand and audit may be preferable to an opaque model whose errors are difficult to explain.

The system architecture is intentionally designed without Java. Python provides the core programming environment for data processing, machine learning, backend development, and database integration. FastAPI can expose secure application programming interfaces between the user interface, data layer, and machine-learning service. HTML, CSS, and JavaScript can provide the web interface, while a mobile application can be added later if actual device integrations are required. SQLite is sufficient for a prototype, while PostgreSQL can support a larger deployment. Pandas and NumPy can process time-series signals, and Scikit-learn can provide standard machine-learning algorithms and evaluation utilities.

The data pipeline begins with explicit consent. A student should understand what categories of information are collected, why they are collected, how long they will be retained, who may receive derived risk information, and how consent can be withdrawn. The system should collect the minimum information necessary for the stated purpose. Raw location traces, message contents, contact names, and other highly identifying information should not be collected merely because a device makes them technically available. Where possible, the system should convert raw signals into coarse features at the point of collection and discard unnecessary raw data.

After consent, the platform performs data-quality checks. Passive sensing is inherently incomplete: a phone may be switched off, a wearable may be removed, permissions may change, a student may leave the device behind, or operating-system updates may interrupt collection. Missingness is therefore a meaningful engineering issue. The proposed system records data coverage and avoids producing strong risk conclusions when the evidence is insufficient. Features may include sleep duration, sleep regularity, activity level, sedentary time, device interaction frequency, mobility regularity, and coarse social-interaction indicators. These features are aggregated over appropriate time windows such as one day, three days, and seven days.

Feature engineering converts raw measurements into interpretable variables. Examples include the change in average sleep duration from the personal baseline, the percentage change in activity, the number of active periods during the day, the regularity of daily routines, and changes in interaction frequency. A rolling baseline can be used so that the model adapts gradually to legitimate changes in the student's schedule. The system should also record confidence or coverage for each feature, because a risk score based on incomplete data should be treated differently from one based on high-quality observations.

The machine-learning component can be trained using labelled wellbeing data. For a student project, a realistic approach is to use a public research dataset where available or a carefully designed synthetic dataset for the software demonstration, with voluntary self-report check-ins used as a prototype label. A real deployment would require a formal research protocol, ethics approval where applicable, clinically appropriate outcome measures, and validation with representative participants. The model should be evaluated using stratified cross-validation and metrics such as precision, recall, F1-score, ROC-AUC, PR-AUC, calibration, and confusion matrices. Because missed high-risk cases may be important, recall should be examined carefully, while excessive false alerts can create alert fatigue and unnecessary anxiety.

The proposed risk engine should combine model output with safety rules rather than relying on a single machine-learning prediction. For example, a high-risk state might require a persistent deviation across several days, sufficient data coverage, and agreement among multiple signals. A rule-based safety layer can prevent a single unusual night or a temporary change in phone use from immediately triggering an escalation. The final output should be described as a wellbeing-risk indicator, not a probability that a student has a psychiatric disorder.

The intervention layer is designed around supportive communication. A student who crosses a threshold can receive a message such as: “We noticed some changes in your recent wellbeing patterns. If you would like, you can complete a short check-in or explore campus support resources.” The message should avoid diagnostic labels and should not imply that the system knows the student's feelings. The student can choose whether to complete the check-in, contact a counsellor, view self-help resources, or take no action. For a genuine emergency, the application should direct the user to the university's established emergency and crisis procedures rather than attempting to act as an emergency response service.

Counsellor workflows should also be carefully designed. Counsellors do not need unrestricted access to raw sensor streams. They can receive a prioritised queue containing a pseudonymous student identifier, risk band, persistence of the signal, data-quality indicator, and relevant high-level changes. If the student has consented to proactive outreach, the counsellor can initiate a confidential check-in through the platform. The interface should record outreach status and prevent multiple staff members from contacting the same student unnecessarily. This creates a human-in-the-loop system in which machine learning supports prioritisation but a trained professional interprets the context and decides what action is appropriate.

Privacy-preserving aggregate analytics form the administrative layer. University administrators may need to understand whether wellbeing indicators are changing across a semester, programme, or broad demographic group, but such analytics do not require individual-level dashboards. The platform can display aggregated percentages, trend lines, and participation rates subject to minimum group-size rules. Small groups should be suppressed or combined to reduce re-identification risk. Differential privacy or related techniques could be investigated in future versions for more advanced aggregate reporting. The underlying principle is that operational usefulness should not require unnecessary disclosure of individual students.

Security is equally important because behavioural data can become sensitive when combined. The system should use HTTPS for data transmission, secure authentication, role-based authorisation, pseudonymous student identifiers, encrypted storage for sensitive fields where appropriate, audit logs, retention policies, and controlled administrator access. Passwords should never be stored in plain text. Tokens should expire and be revocable. Database backups should be protected. Development and production environments should be separated. The prototype should also include a data-deletion or consent-withdrawal workflow so that a participant can stop contributing data and, according to the institution's policy, request deletion of stored information.

Fairness and bias must be treated as engineering requirements rather than optional features. Passive sensing performance may vary with device type, operating system, battery settings, accessibility configurations, living arrangements, cultural practices, work schedules, and socioeconomic circumstances. A model trained on a small or homogeneous sample may not generalise to the broader campus population. Therefore, the evaluation should examine performance across relevant groups when sample size permits and should report limitations rather than claiming universal accuracy. The system should never use a risk score to punish students, make academic decisions, restrict services, or label individuals as mentally ill.

The project also requires a clear distinction between early warning and diagnosis. The purpose is to identify behavioural deterioration that may justify a supportive check-in, not to determine whether a student meets diagnostic criteria. A diagnostic claim would require clinical assessment and appropriate validation. This distinction affects the user interface, model labels, documentation, consent language, and evaluation criteria. The project can therefore be positioned as a wellbeing-support and triage-assistance platform rather than a medical diagnostic product.

From a software-engineering perspective, the project can be developed incrementally. The first version can use simulated data and demonstrate the full workflow from consent to risk calculation and dashboards. The second version can introduce a machine-learning model trained on a suitable research dataset. The third

version can add stronger security, audit logging, aggregate analytics, and usability testing. Actual device integrations should be treated as a later module because mobile operating systems and wearable platforms impose permissions, battery, privacy, and data-access constraints.

The expected contribution of the project is therefore a practical demonstration of how machine learning and passive sensing can be integrated into a campus support workflow while keeping consent, privacy, uncertainty, and human oversight central. The literature shows promise but also highlights small samples, missing data, limited generalisability, privacy concerns, and the need for clinical integration. The proposed system responds to these gaps by combining personal-baseline modelling, data-quality awareness, configurable thresholds, privacy-preserving analytics, and human-in-the-loop intervention. The final prototype should demonstrate not simply that a classifier can predict a label, but that a complete and ethically constrained system can translate uncertain behavioural signals into a cautious and supportive next step.

2.1 Proposed System Architecture

Student / Device	Backend	Analytics	Human Support
Consent Passive / simulated signals Student check-in	FastAPI Authentication Data validation SQLite/PostgreSQL	Pandas/NumPy Personal baseline ML model Risk engine	Student message Counsellor queue Aggregate admin analytics

Important scope: The proposed platform is a decision-support and wellbeing-support prototype. It does not diagnose mental illness, replace counselling, or independently determine that a student is in a psychiatric crisis.

3. PROBLEM ANALYSIS / LITERATURE REVIEW

The literature indicates that passive sensing can capture behavioural features related to sleep, activity, mobility, device use and social interaction, and that machine-learning methods can model associations with mental-health outcomes. At the same time, the evidence base has recurring limitations: many studies use small samples, short observation periods, heterogeneous sensors, incomplete data, and research-specific applications. These limitations mean that a campus early-warning system should not be presented as a clinically validated diagnostic device. The literature instead supports a cautious research direction in which behavioural changes are used to trigger supportive attention.

A second major theme is privacy. Smartphone and wearable data can reveal much more than a participant may expect, particularly when location, communication, and behavioural traces are combined. Ethical work on digital phenotyping consistently emphasises informed consent, transparency, accountability, fairness and data protection. Therefore, the proposed system deliberately avoids collecting message contents, contact names and detailed location histories. The system uses derived features and pseudonymous identifiers wherever possible, while exposing only minimal information to counsellors and aggregated information to administrators.

A third theme is personalisation. A population-level model can mistakenly interpret legitimate individual differences as risk. Longitudinal research with students shows that behaviour changes across the academic calendar and in response to major events. A personal baseline therefore provides a useful design principle: the system should focus on persistent deviation from an individual's recent pattern and should incorporate data-quality measures before generating an alert.

The following review summarises more than 20 relevant papers and research directions that inform the proposed project.

3.1 Wang et al. (2014) — StudentLife

The StudentLife project demonstrated continuous smartphone sensing among college students and linked objective measures of sleep, activity, sociability and phone-derived behaviour with stress, mood, wellbeing and academic outcomes. It provides one of the strongest conceptual foundations for a campus-oriented sensing platform and supports the use of behavioural signals as contextual indicators rather than standalone diagnoses.

3.2 Saeb et al. (2015) — Mobile phone sensor correlates of depressive symptom severity

This exploratory study examined smartphone sensor measures in daily life and found associations between passive behavioural patterns and depressive symptom severity. It supports the feasibility of deriving mental-health-relevant features from routine phone data, while also illustrating the need for larger and more diverse validation samples.

3.3 Canzian and Musolesi (2015) — Mobility trajectories and depression

This work investigated unobtrusive monitoring of depressive states using smartphone mobility traces. Reduced or altered mobility can be informative, but location information is highly sensitive. The proposed project therefore uses coarse mobility-derived features rather than storing detailed location histories.

3.4 Ben-Zeev et al. (2016) — Mobile behavioral sensing in schizophrenia

The study demonstrated the feasibility and acceptability of multimodal smartphone sensing using accelerometer, microphone, GPS, Wi-Fi and Bluetooth information. Privacy concerns were also reported by some participants, reinforcing the need for explicit consent and minimisation in the proposed campus

system.

3.5 Onnela and Rauch (2016) — Smartphone digital phenotyping

This perspective established the broader concept of smartphone-generated digital fingerprints as a potential source of dynamic behavioural markers for mental health. It motivates the use of repeated measurements and personalised temporal patterns in the proposed system.

3.6 Torous et al. (2016) — New tools for psychiatric research

The authors described scalable and customisable smartphone research platforms. Their work is relevant to the proposed modular architecture, in which sensing, analysis, feedback and research evaluation can be separated into components.

3.7 Saeb et al. (2016) — Location sensor data and depressive symptoms

This study evaluated relationships between mobile phone location sensor information and depressive symptom severity. The findings support mobility as a potentially useful feature while highlighting the privacy implications of fine-grained location data.

3.8 Cornet and Holden (2018) — Systematic review of passive sensing

A systematic review identified 35 empirical studies and found that accelerometry, location, audio and usage data were among the most common sensing streams. The review also identified technical, methodological and privacy challenges. These findings directly inform the proposed emphasis on data quality, privacy and problem-specific feature selection.

3.9 Insel (2018) — Digital phenotyping for psychiatry

Insel described digital phenotyping as a potential global tool for psychiatry, emphasising the opportunity to move from episodic assessments toward more continuous behavioural information. The proposed system adopts the monitoring concept but deliberately avoids presenting itself as a diagnostic system.

3.10 Huckins et al. (2020) — College students during COVID-19

Longitudinal mobile sensing and ecological momentary assessment among college students showed that major contextual changes were accompanied by changes in behaviour and mental health. This supports longitudinal modelling and demonstrates why temporal context matters when interpreting behavioural signals.

3.11 Jacobson et al. (2020) — Passive sensing of moment-to-moment depressed mood

The study used smartphone sensing and machine learning to predict future depressed mood within a day in a clinically depressed sample. It supports the technical possibility of short-horizon prediction but also illustrates that predictive models require careful validation before being translated into interventions.

3.12 D'Alfonso et al. (2021) / ethical digital phenotyping Delphi

The Delphi study identified privacy, transparency, consent, accountability and fairness as central ethical issues. These principles are incorporated into the proposed architecture through opt-in participation, role-based access, data minimisation and explicit communication of limitations.

3.13 Torous et al. / college digital phenotyping review

A clinical review of digital phenotyping studies involving college students found that location, accelerometer and social information were common data streams and that sleep, exercise and social interaction were

recurring behavioural constructs. This closely matches the proposed feature set.

3.14 GLOBEM-related college sensing research

Longitudinal college sensing work has shown that phone-use and behavioural features can be associated with anxiety and other wellbeing outcomes. Such studies reinforce the value of temporal windows and individual trajectories rather than one-time snapshots.

3.15 Nepal et al. (2024) — Four-year college mobile sensing

A four-year mobile sensing study followed more than 200 undergraduate students and demonstrated the value of long-term sensing for understanding resilience, behaviour and mental-health changes. It also highlighted practical issues such as operating-system updates, missing data and long-term data completeness.

3.16 AWARE-based sensing studies

The AWARE platform provides a framework for collecting mobile context data and has supported research using screen activity and other passive signals. The proposed system can use the same conceptual separation between data collection, feature extraction and analysis while limiting the raw information collected.

3.17 Feinberg et al. (2018) — iSee user needs

A qualitative study explored user and clinician needs for a mobile behavioural-sensing technology for depression management. The work is relevant because technical accuracy alone is insufficient: the system must provide understandable information and feedback that fits counselling workflows.

3.18 BiAffect study (2018) — keyboard metadata

Research on passive keyboard metadata demonstrated that interaction patterns can be used to build digital phenotypes related to mood disturbance. The project does not require keyboard-content collection; instead, it takes the privacy-preserving lesson that derived interaction statistics may be preferable to sensitive raw content.

3.19 Passive smartphone risk stratification research

Studies combining passive smartphone features with clinical or self-report outcomes suggest that behavioural sensing can contribute to risk stratification. However, these approaches should complement rather than replace clinical assessment, which is consistent with the proposed human-in-the-loop design.

3.20 2024 systematic review of digital phenotyping for stress, anxiety and mild depression

A recent systematic review identified 36 papers covering 40 studies and found recurring use of GPS, Bluetooth, accelerometer, microphone, gyroscope, Wi-Fi and phone-use features. Most included studies used machine learning. The review also highlights heterogeneity and methodological challenges, supporting cautious model evaluation.

3.21 2025 college passive-sensing study using LightGBM

A recent longitudinal pilot study of college students combined wearable and smartphone sensing and used LightGBM to detect depressive symptoms. Sleep quality and missed mobile interactions were among the most predictive features. The small sample size demonstrates both the promise of multimodal sensing and the need for broader validation.

3.22 2026 Korean community digital phenotyping study

Recent work combining passive smartphone sensing with brief self-report measures reported useful predictive performance for high-risk depression and anxiety and identified mobility, activity regularity and sleep-related behaviours as informative. The study supports a multimodal approach while showing that passive and active data can complement one another.

3.24 Research Gap Identified

The literature supports passive sensing and machine learning, but a practical campus system must connect prediction to a complete, privacy-conscious intervention workflow. The main gap addressed by this project is therefore system integration: consent management, personalised baselines, data-quality checks, interpretable risk scoring, supportive student messaging, counsellor prioritisation, privacy-preserving aggregate analytics, and security are designed as one end-to-end platform. The project also emphasises that an alert is an invitation for human support, not a clinical diagnosis.

4. PROPOSED METHODOLOGY

The project will follow an iterative, modular methodology combining software engineering, data analytics, machine learning and privacy-by-design. Development will begin with simulated or voluntarily entered data so that the full workflow can be demonstrated without requiring unrestricted access to real student device data. Actual device integrations can be added later as a separate module after appropriate permissions and institutional approvals.

1. Requirement and Ethical Analysis

Define users, use cases, data categories, consent requirements, intervention rules and institutional roles. The system will explicitly state that it provides wellbeing-risk indicators rather than psychiatric diagnoses. A data inventory will identify what is collected, why it is needed, who can access it, and how long it is retained.

2. Consent and Participant Management

Create a consent screen explaining data categories, purpose, expected benefits, limitations, sharing, withdrawal and deletion options. Participation is opt-in. A consent table stores status and timestamps without storing unnecessary sensitive information.

3. Data Acquisition

For the prototype, data can be simulated or voluntarily entered. The design will support future connectors for smartphone or wearable data. Candidate signals include sleep duration/regularity, activity level, device interaction frequency, mobility regularity and coarse social-interaction frequency. Raw message content, contact names and detailed communications are excluded.

4. Data Preprocessing

Use Python, Pandas and NumPy to clean timestamps, handle duplicates, detect impossible values, calculate coverage, and normalise numerical features. Missingness is retained as a quality indicator. Features with insufficient coverage are not allowed to drive a high-confidence alert.

5. Personal Baseline and Feature Engineering

For each consenting participant, compute rolling personal baselines. Derive features such as sleep deviation, activity deviation, social-interaction change, routine irregularity, device-use change and multi-day persistence. Use one-day, three-day and seven-day windows where appropriate.

6. Machine-Learning Model Development

Train interpretable baseline models such as Logistic Regression and Random Forest. If the dataset supports it, compare Gradient Boosting. Use train/validation/test separation, stratified cross-validation, class-imbalance handling, and calibration. Evaluate precision, recall, F1-score, ROC-AUC and PR-AUC. Prefer models that are stable and explainable.

7. Risk Engine and Safety Layer

Combine model output with data-quality rules and persistence rules. Example risk bands can be Low (0–30), Moderate (31–60) and High (61–100), but thresholds will be treated as configurable prototype parameters rather than clinically validated cut-offs. A single abnormal day should not automatically create an escalation.

8. Student Check-in Module

When an alert condition is met, send a non-diagnostic and supportive message. The student can complete a brief check-in, view campus resources, request human contact, or dismiss the prompt. Emergency messaging will point to established university crisis procedures rather than pretending that the software can provide emergency intervention.

9. Counsellor Dashboard

Provide a prioritised queue based on risk band, persistence and data quality. Show only the minimum information necessary for outreach. Counsellors can record outreach status and follow-up. The system should not expose raw sensor streams by default.

10. Privacy-Preserving Administration

Provide aggregate campus analytics with minimum group-size rules. Suppress small cells and avoid unnecessary demographic slicing. Future research can add differential privacy or federated learning, but the first prototype will prioritise straightforward data minimisation and pseudonymisation.

11. Security and Access Control

Implement authenticated access, role-based permissions, secure sessions/tokens, HTTPS in deployment, password hashing, protected database access, audit logging, and retention/deletion controls. Student, counsellor and administrator roles will have different permissions.

12. Testing and Evaluation

Conduct unit testing, API testing, model testing, usability testing and security checks. Evaluate false positives and false negatives, alert frequency, latency, missing-data robustness and user comprehension. A successful prototype is one that completes the end-to-end workflow reliably and transparently, not one that merely maximises a single ML metric.

13. Deployment Strategy

Develop locally in VS Code. Use FastAPI as the backend, HTML/CSS/JavaScript for the interface, SQLite for the prototype database and PostgreSQL as the scalable alternative. The machine-learning model is loaded by the Python service and invoked through the risk-analysis pipeline.

4.14 Technology Stack

Layer	Technology	Purpose
Programming	Python	Core backend, data processing and ML
Backend/API	FastAPI	Secure REST APIs and application logic
Frontend	HTML, CSS, JavaScript	Student and counsellor interfaces
Data	SQLite / PostgreSQL	Prototype and scalable relational storage
ML	Scikit-learn	Classification, evaluation and preprocessing
Analytics	Pandas, NumPy	Cleaning, aggregation and feature engineering
Charts	Chart.js	Dashboard visualisation
Development	VS Code, Git	Implementation and version control

4.15 Proposed Data Flow

1	Consent	Student opts in and selects permitted data categories.
2	Collection	Passive/simulated signals are received through the API.
3	Quality	Coverage, missingness and validity are checked.
4	Features	Rolling baseline deviations and time-window features are generated.
5	Prediction	ML model produces a wellbeing-risk estimate.
6	Safety rules	Persistence and confidence rules reduce single-event false alerts.
7	Support	Student receives a check-in; counsellor receives a priority item if permitted.
8	Analytics	Administrators see only approved aggregate trends.

5. OBJECTIVES

1. To design an ethical, opt-in platform for early identification of behavioural changes associated with declining student wellbeing.

2. To collect and process selected passive or simulated signals such as sleep, activity, device-use and social-interaction frequency while following data-minimisation principles.

3. To develop a personalised baseline method so that risk is assessed primarily from meaningful changes in an individual student's routine.

4. To compare interpretable machine-learning approaches for generating a wellbeing-risk indicator from longitudinal behavioural features.

5. To implement a supportive, confidential student check-in workflow that avoids diagnostic language.

6. To provide counsellors with a prioritised outreach queue using only the minimum information necessary for proactive support.

7. To provide administrators with privacy-preserving aggregate analytics rather than unnecessary individual-level information.

8. To implement consent management, role-based access, pseudonymisation, security controls, audit logging and data-retention mechanisms.

9. To evaluate the prototype for predictive performance, data quality, usability, fairness, privacy and robustness to missing data.

10. To demonstrate a complete Python-based implementation without requiring Java, using FastAPI, HTML/CSS/JavaScript, SQLite/PostgreSQL and Scikit-learn.

5.1 Expected Outcomes

1. Working student consent and profile module.

2. Simulated or voluntarily entered wellbeing-signal ingestion pipeline.

3. Python preprocessing and personal-baseline feature engineering pipeline.

4. Machine-learning risk-scoring module with evaluation metrics.

5. Student wellbeing check-in interface.

6. Counsellor prioritisation dashboard with role-based access.

7. Privacy-preserving aggregate dashboard.

8. Secure database schema with consent, risk and outreach records.

9. Technical documentation describing architecture, limitations and ethical safeguards.

5.2 Scope and Limitations

The first version is a prototype intended for academic demonstration. It will not be used to make disciplinary, academic, employment, insurance, or admission decisions. It will not provide psychiatric diagnosis or emergency response. Device-level integrations may be simulated if appropriate APIs or permissions are not available. Model performance will depend on the quality and representativeness of the available training data. Any real-world deployment would require additional clinical validation, institutional governance, legal/privacy review, security assessment, user research, and appropriate ethical approval.

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